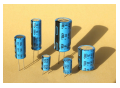
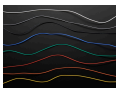
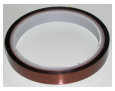
Starling 2 Max — shopping list

Quantities cover **both** drones
Each item has its own card on the following pages

Sim Lim Tower — 10 Jalan Besar · components · most shops close ~7:30 pm

		ITEM	QTY	KEY SPEC
☐		**1. Addressable LED strip**	1 metre	WS2812B, 5 V, 60 LED/m, ≤ 10 mm wide, bare, 3 wires
☐		**2. Cooling fan (+ backup blower)**	2 square fans + 1 blower	30×30×10 mm, 5 V, 2 wires, ≤ 0.15 A; + 3010 blower
☐		**5. Resistors**	10	330 Ω (or 470 Ω), 1/4 W
☐		**6. Capacitors**	4 × 470 µF + 4 × 100 nF	470 µF 10–16 V electrolytic + 100 nF ceramic
☐		**7. Silicone hook-up wire**	~2 m each, 3 colours	26 AWG silicone, stranded, 3 colours
☐		**8. Kapton tape + heat-shrink**	1 roll + assortment + 20 cm clear	Kapton ~10 mm; heat-shrink 1.5–6 mm + 15–20 mm clear
☐		**4. Level-shifter chip**	2 (+1 spare)	74HCT125, DIP-14
☐		**3. 5 V step-down (buck) module**	2 (+1 spare)	MP1584 (or Mini-360), 5 V out, ≥ 1 A

Sim Lim Square — 1 Rochor Canal Road · hobby / FPV · next door

		ITEM	QTY	KEY SPEC
☐		**9. Propellers**	at least 2 sets of 4	Gemfan 7037-3: 7 in, 3.7 pitch, 3 blades, 5 mm hole

Bring

This sheet (prices go on the **price log**, page 8) · the **printed prop gauge** and **30 × 30 × 10 fan block** · a ruler or calipers

Every purchase

Ask for an **itemised receipt** with the shop's name and GST number. Compare two or three shops before buying.

Why a buck module and a chip?

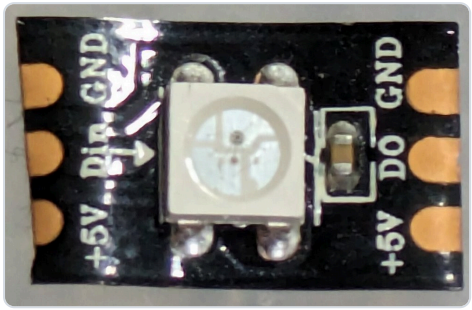
The LED strip and fan get their own 5 V supply (item 3), so a wiring fault on an arm can't take down the flight computer. The chip (item 4) makes the LED data signal strong enough for a 5 V strip.

1 · LED strip

1 Addressable LED strip Sim Lim Tower

Buy: 1 metre

covers both drones (2 × 16 LEDs) with spares



One bare WS2812B segment: 3 pads at each cut (+5V, Din/DO, GND)

Photo: John McMaster, Wikimedia Commons, CC BY 4.0

ASK FOR

“Do you have WS2812B addressable LED strip, 5 volt, 60 LEDs per metre, the narrow bare kind without the silicone sleeve? The newest version, V5, if you have it.”

SPEC	MUST BE	WHY
Type	Individually addressable, single data wire: **WS2812B** (or SK6812)	The drone sends one data signal that sets each LED separately
Wires	**3**: +5 V, data, ground	4-wire strips need a clock line the drone doesn't have
Voltage	**5 V**	12 V strips won't light
LED spacing	**60 LEDs per metre** (16.7 mm apart)	4 LEDs fit on each ~75 mm arm
Width	**10 mm or less**; 5 mm “narrow” is ideal	The arms are about 10 mm wide
Finish	**Bare** (IP30), no silicone sleeve or tube	A sleeve makes it ~14.5 mm wide and heavier
Colours	RGB is fine; RGBW (SK6812) also works	Software handles both
Chip version	**WS2812B V5** preferred, if the shop knows	Accepts the drone's 3.3 V data even on a 5 V supply

✗ **Do not buy:** APA102, DotStar, SK9822 or WS2801 (4 wires) · WS2815 (12 V) · Non-addressable RGB strip (whole strip one colour) · Silicone-sleeved or tube “waterproof” strip · 30 LEDs per metre (only 2 fit per arm)

✓ **Check:** Hold the strip against a ruler: 10 LEDs should span ~167 mm, and the width must be ≤ 10 mm.

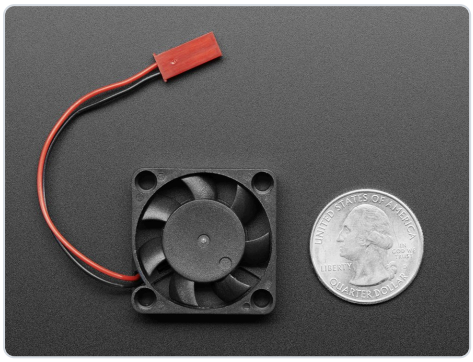
2 · Cooling fan

2 Cooling fan (+ backup blower)

Sim Lim Tower

Buy: 2 square fans + 1 blower

one fan per drone; the blower is a backup if the shroud can't feed the gaps



Shape shown is 8 mm thick: buy 30 × 30 × 10 mm

Photo: Adafruit (product 3368)



Backup blower: side outlet. Must be 30 × 30, 5 V, 2 wires

Photo: The Pi Hut

ASK FOR

“A 30 by 30 by 10 millimetre DC fan, 5 volt, two wires only. And a 30 millimetre blower fan, 5 volt, also two wires.”

SPEC	MUST BE	WHY
Size	**30 × 30 × 10 mm** (“3010”)	Fits the 36 mm-wide board stack; the 10 mm-thick version pushes harder through the 2 mm gaps
Voltage	**5 V DC**	The drone's fan port gives 5 V; a 12 V fan barely turns
Wires	**Exactly 2**: red (+) and black (−)	Speed is controlled by switching the ground; 3- and 4-wire fans stutter or won't start
Current	**0.15 A (150 mA) or less** on the label	The fan port is limited to ~0.4 A including start-up
Bearing	**Ball** or fluid / hydraulic (“FDB”)	It will vibrate; plain sleeve bearings wear out
Plug	Any	Cut off: the fan is soldered to the buck module (item 3)
Backup blower	**30 × 30 × 10 mm blower** (“snail shell”, side outlet), 5 V, 2 wires, ≤ 0.15 A	Pushes harder into a single narrow gap
Reference part	Sunon MF30100V1-1000U-A99 (5.5 CFM, 0.12 A)	Any brand matching the rows above is fine

✗ **Do not buy:** 12 V or 24 V fans · 3-wire (tacho) or 4-wire (PWM) fans · 25 mm or 40 mm sizes · Sleeve-bearing fans

✓ **Check:** Compare against your printed 30 × 30 × 10 mm reference block.

3–4 · Buck module and level shifter

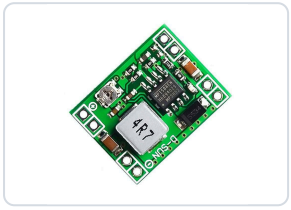
How these are used: the buck module turns the battery's 12–16.8 V into 5 V for the LED strip and the fan, separately from the flight computer. The 74HCT125 lifts the drone's 3.3 V LED data signal to 5 V.

3 5 V step-down (buck) module

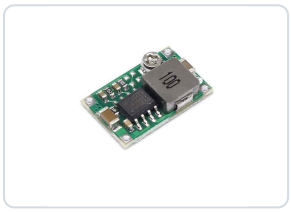
Sim Lim Tower

Buy: 2 (+1 spare)

one per drone: powers the LED strip and the fan



MP1584EN (~22 × 17 mm)
Photo: Kunkune



Mini-360 (~17 × 11 mm)
Photo: Kunkune

ASK FOR
“An MP1584 step-down buck converter module. A Mini-360 is also fine.”

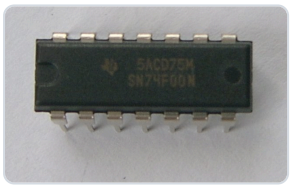
SPEC	MUST BE	WHY
Output	**5.0 V**, **1 A or more** (MP1584 ~3 A)	Powers the LED strip and fan, separately from the flight computer
Input	Accepts **≥ 17 V** (MP1584 28 V, Mini-360 23 V max)	Runs from the 4S battery, up to 16.8 V
Size	MP1584 ~22 × 17 mm; Mini-360 ~17 × 11 mm	Light; mounts at an arm root
Adjustable?	Trim screw: **set 5.0 V with a multimeter** before use	Often shipped at another voltage

✗ **Do not buy:** Linear 7805 regulators (would burn ~6 W as heat) · Big LM2596 boards (heavy)

4 Level-shifter chip

Sim Lim Tower

Buy: 2 (+1 spare)



Shown: a 74-series DIP-14 chip. Buy 74HCT125
Photo: Stefan Riepl, Wikimedia Commons, public domain

ASK FOR
“74HCT125, the quad buffer chip, in DIP-14.” (The letter T matters.)

SPEC	MUST BE	WHY
Part	**74HCT125** (or 74AHCT125)	Raises the drone's 3.3 V LED data signal to 5 V for the strip
Package	DIP-14 (through-hole)	Easy to hand-solder onto a scrap of perfboard
With it	a **100 nF** ceramic capacitor (item 6) and a small perfboard scrap	Standard decoupling for logic chips; keeps the assembly tidy

✗ **Do not buy:** 74HC125 or 74LS125 (wrong input levels)

✓ **Check:** If there's no 74HCT125 anywhere: RS Singapore / element14 Singapore stock it online.

5–6 · Resistors and capacitors

5 Resistors Sim Lim Tower

Buy: 10

they cost cents



330 Ω: orange-orange-brown

Photo: oomlout, Wikimedia Commons, CC BY-SA 2.0

ASK FOR

“330 ohm resistors, quarter watt.” (470 ohm is also fine.)

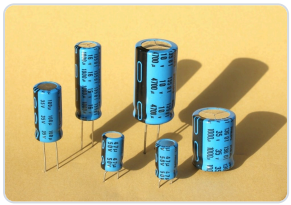
SPEC	MUST BE	WHY
Value	**330 Ω** (bands orange-orange-brown) or **470 Ω** (yellow-violet-brown)	Protects the LED data line; anything 300–500 Ω works
Rating	1/4 W, carbon or metal film, through-hole	Ordinary, smallest common size

✗ **Do not buy:** SMD chip resistors (too small to hand-solder onto wires)

6 Capacitors Sim Lim Tower

Buy: 4 × 470 µF + 4 × 100 nF

one per strip + spares



Various sizes shown: buy 470 µF, 10–16 V. Stripe = negative leg

Photo: Elcap, Wikimedia Commons, CC0

ASK FOR

“470 microfarad electrolytic capacitor, 10 or 16 volt, the smallest size you have. And some 100 nanofarad ceramic capacitors, code 104.”

SPEC	MUST BE	WHY
Value	**470 µF**	Smooths the LED strip's power where it starts
Voltage	**10 V or 16 V** (6.3 V is the minimum)	Strip runs at 3.3 or 5 V
Type	Radial aluminium electrolytic, small (≤ 8 mm wide)	Light; the stripe marks the negative leg
Also	**100 nF ceramic** (marked “104”), ×4	Sits across the 74HCT125's supply pins

✗ **Do not buy:** Large 25 V+ cans (heavier than needed) · Ceramic capacitors (much smaller value)

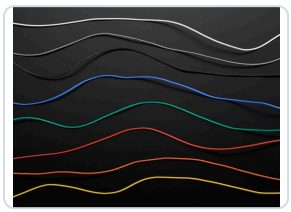
7–8 · Wire, tape, heat-shrink

7 Silicone hook-up wire

Sim Lim Tower

Buy: ~2 m each, 3 colours

red, black and one other colour



26 AWG silicone stranded wire
Photo: Adafruit (product 1970)

ASK FOR
“26 AWG silicone wire, stranded, in red, black and one other colour.”

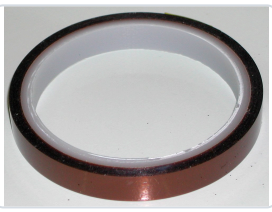
SPEC	MUST BE	WHY
Gauge	**26 AWG**	Enough for the LED current; thin enough for the ESC's small solder pads
Insulation	**Silicone**	Doesn't melt back while soldering; flexes with vibration
Core	**Stranded** (many fine strands)	Solid core cracks with vibration

✗ **Do not buy:** PVC hook-up wire · Solid-core wire

8 Kapton tape + heat-shrink

Sim Lim Tower

Buy: 1 roll + assortment + 20 cm clear



Kapton tape: amber, thin
Photo: Jérôme, Wikimedia Commons, CC BY-SA 3.0

ASK FOR
“Kapton tape, the amber polyimide kind, about 10 millimetres wide. A small heat-shrink assortment, 1.5 to 6 millimetres, and a short length of clear 15 to 20 millimetre heat-shrink.”

SPEC	MUST BE	WHY
Kapton width	**~10 mm** (6–12 mm OK), polyimide, amber	Wraps the strip ends onto the 10 mm arms; heat-proof, holds under vibration
Heat-shrink	**1.5–6 mm** assortment, plus **15–20 mm clear**	Small: solder joints. Large: insulates the buck module and chip, since carbon fibre conducts

✗ **Do not buy:** Electrical PVC tape (adhesive creeps) · Heat-shrink in large sizes only (> 10 mm)

9 · Propellers

9 Propellers Sim Lim Square

Buy: at least 2 sets of 4

each set is 2 clockwise + 2 counter-clockwise; props are consumables



The bag to look for: Gemfan 7037-3
Photo: OddityRC, Gemfan 7037-3 listing



7037-3: three blades, 5 mm centre hole
Photo: OddityRC, Gemfan 7037-3 listing

ASK FOR
“Gemfan 7037, three-blade, 7 inch, 5 millimetre centre hole.”

SPEC	MUST BE	WHY
Diameter	**7 inch** (178 mm)	Same as the drone's stock props
Pitch	**3.7** (3.5 acceptable) — **never higher**	Higher pitch overheats the small motors and the ESC
Blades	**3**	Matches stock
Centre hole	**5 mm** (M5)	Fits the motor shaft
Hub	~7 mm thick	Fits the prop nut
Rotation	Set must include both **CW and CCW**	Two motors spin each way
Label	“7037-3” or “7×3.7×3”	First number = diameter, second = pitch

✗ **Do not buy:** 7040, 7042, 7045 or any “freestyle” prop (pitch too high) · Two-blade props (e.g. 7035-2) · 6 or 8 inch props · Props without a 5 mm centre hole

✓ **Check:** Lay it on your printed prop gauge: tips reach the outer ring; the hole fits the gauge's 5 mm hub.

Genuine ModalAI part: MRP-D0012-1-00 (US$29.99 / 4), online only.

Price log

Compare two or three shops before buying

ITEM · QTY	SHOP / UNIT	EXACT ITEM SEEN	PRICE S$	BOUGHT?
1 Addressable LED strip 1 metre				
2 Cooling fan (+ backup blower) 2 square fans + 1 blower				
5 Resistors 10				
6 Capacitors 4 × 470 µF + 4 × 100 nF				
7 Silicone hook-up wire ~2 m each, 3 colours				
8 Kapton tape + heat-shrink 1 roll + assortment + 20 cm clear				
4 Level-shifter chip 2 (+1 spare)				
3 5 V step-down (buck) module 2 (+1 spare)				
9 Propellers at least 2 sets of 4				

Where to go

Call ahead: stock and unit numbers change

Sim Lim Tower — 10 Jalan Besar — electronic components

SHOP	UNIT	TRY HERE FOR
J.C.W. Electronic	—	Fans (item 2) — **call first**; LED strip
Sun-Light Electronics	#03-09	WS2812B LED strip (item 1)
Davjones Technology	#B1-47/11	Buck modules, 74HCT125 (items 3–4)
Continental Electronic Enterprise	#B1-25	Resistors, capacitors, wire, tape, heat-shrink (items 5–8)

From a shop directory about two years old. No online listing confirms the fans, blower or 74HCT125 in stock anywhere in Sim Lim — call first.

Sim Lim Square — 1 Rochor Canal Road — hobby / FPV

SHOP	UNIT	TRY HERE FOR
Hobby Square	#02-60	**Propellers (item 9)**; also silicone wire (item 7). Listed hours Mon–Sat 12:30–7:30 pm

If Hobby Square has no 7037 props

SHOP	WHERE	NOTES
Sky Hobbies	Jalan Sultan	skyhobbies.com.sg · large HQProp stockist
Helicopter Smartfly	Queensway Shopping Centre	smartfly.com.sg · Mon–Sat 12–7 pm, Sun 3–7 pm, closed Wednesday
Rotor Hobby	Kitchener Complex, French Road	shop.rotor.com.sg · +65 6296 3686
OddityRC	Paya Lebar Square (pickup)	oddityrc.com · stocks Gemfan 7037-3 online; allow 5–8 days

Other fallbacks

Fans	Amicus, 8 Ubi Road 2 #03-13 Zervex · +65 6980 6783 · 5 V fans by order; call first
Online, Singapore	Cytron SG and Kuriosity.sg (LED strip, buck modules) · RS Singapore and element14 Singapore (74HCT125, Sunon fans; fast delivery) · DigiKey (a few days)
Online, blower	3D-printer parts sellers and local Shopee sellers (“3010 blower 5V”)